Problem Description | Solution | Video | Comments 0 | Playground

Calculate Mean by Row or Column

Easy | Linear Algebra

Write a Python function that calculates the mean of a matrix either by row or by column, based on a given mode. The function should take a matrix (list of lists) and a mode ('row' or 'column') as input and return a list of means according to the specified mode.

Example:

Input:

```
matrix = [[1, 2, 3], [4, 5, 6], [7, 8, 9]], mode = 'column'
```

Output:

```
[4.0, 5.0, 6.0]
```

Reasoning:

Calculating the mean of each column results in [(1+4+7)/3, (2+5+8)/3, (3+6+9)/3].

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Contributors:

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StoatScript

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Notebook Mode | NumPy

```python
# This needs to be imported at the top of the file, inside the Deep-ML code writing section.
import numpy as np


def calculate_matrix_mean(matrix: list[list[float]], mode: str) -> list[float]:

    # Initially converting the python list of lists into a numpy array. And at the same time changing the data type of the array into float.
    matrix = np.array(matrix).astype(float)

    # Calculating the mean of the matrix based on the given mode.
    if mode == 'row':
        means = np.mean(matrix, axis=1)
    else:
        means = np.mean(matrix, axis=0)

    # Now converting the numpy array into a list of floats.
    means = means.tolist()

    # Returning the list of means.
    return means
```

Run Code | Submit Code | Reset | Save Code | Submissions 3

Case 1 +

Test Code

```
print(calculate_matrix_mean([[1, 2, 3], [4, 5, 6], [7, 8, 9]], 'column'))
```

Expected Output

```
[4.0, 5.0, 6.0]
```

Test Results 4/4

Test 1 | Test 2 | Test 3 | Test 4

Test Case

```
print(calculate_matrix_mean([[1, 2, 3], [4, 5, 6], [7, 8, 9]], 'column'))
```

Expected

```
[4.0, 5.0, 6.0]
```

Your Output — Passed

```
[4.0, 5.0, 6.0]
```

Ask Tutor